

DPCN Assignment 1: Opinion Network Formation from an Ordinal Attitude Questionnaire

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Abstract

We construct and analyse an opinion network from a 60-item ordinal attitude questionnaire answered by 96 respondents across four thematic blocks (technology, education, society, environment). The work has three parts. First, we sanitise the data while preserving the distinction between two different missingness mechanisms: structural blanks left when a respondent abandons the survey, and scattered “No Comments” (NC) responses inside otherwise complete questionnaires. NC is not recoded as Neutral and NC users are not discarded; missing responses are instead recovered by question-wise regularised proportional-odds ordinal logistic regression, validated by artificial masking under both missingness patterns. Second, we show that the response distribution is severely ceilinged—79.8% of valid answers fall on the agree side and the grand mean is +1.14 on a $[-2, +2]$ scale—so that raw similarity measures how agreeable a respondent is rather than what they believe. Row-centring each respondent removes this component and reduces the number of item pairs correlated at $|r| \geq 0.30$ from 327 to 42. Third, we build correlation networks over the sanitised matrix and run Deffuant–Weisbuch bounded-confidence dynamics on a respondent network. We show analytically and by controlled experiment that the widely reported cluster plateau at small confidence thresholds is an arithmetic artefact of the five-point response grid, while the difference between the sparse network and a well-mixed population is a genuine effect of topology.

1 Introduction

Likert-scale questionnaires are a natural setting for studying how attitudes in different domains relate to one another. Although each item is answered independently, responses exhibit substantial dependence

across questions and across thematic domains. Network representations model these dependencies directly: questionnaire items (or respondents) become vertices, and statistical association becomes weighted edges.

Turning a questionnaire into a network is not, however, a mechanical step. Three decisions determine whether the resulting graph carries signal or merely re-describes the response format.

How missing data is handled. Missing responses arise both from respondents stopping before reaching later questions and from respondents declining to answer a particular item. These are different mechanisms [8] and should not be silently merged, and assigning a numerical value such as Neutral to a declined answer injects information the respondent never supplied.

Whether response style is removed. When respondents agree with almost everything, correlation between any two items is inflated by a shared “agreeableness” component rather than genuine item association [7]. Unless this is removed, the network largely encodes response style.

What the network is compared against. A correlation-derived network has transitivity built into its construction, so clustering and modularity appear even in structureless data. Any structural claim needs a null model beside it.

This report addresses all three. Section 2 documents the dataset and its ceiling effect, Section 3 the sanitisation, imputation and row-centring pipeline, Section 4 the individual analyses and visualisations, and Sections 5–6 the findings and their limits.

Team and code. Team [TEAM NAME]. All code, figures, and the data pipeline are available at https://github.com/shreyash-lohare/dpcn_a1_opinion_network. The full analysis is reproducible from a fresh clone via `python run_all.py` followed by the figure modules in `src/`.

2 Dataset Documentation

The questionnaire contains 60 items in four thematic blocks of 15: technology (T01–T15), education (E01–E15), society and ethics (S01–S15), and environment (V01–V15). Each item offers five ordered categories plus an explicit No Comments option. For numerical

analysis the categories are encoded symmetrically,

$$SD = -2, \quad D = -1, \quad N = 0, \quad A = +1, \quad SA = +2,$$

which represents the ordering of the categories without asserting that adjacent categories are equally spaced. Descriptive statistics in this section are reported on the original 1–5 scale (subtract 3 to recover the encoding above).

The raw file contains 96 respondent records and $96 \times 60 = 5,760$ cells, of which 5,255 were answered and 5,218 carry a substantive ordinal response. Blocks are internally consistent to varying degrees: Cronbach’s α [2] is 0.87 for environment and 0.82 for society, but only 0.65 for technology and 0.58 for education, and 0.885 across all 60 items.

2.1 The response ceiling

The single most consequential property of this dataset is its ceiling. Figure 1(a) shows that 79.8% of valid answers are Agree or Strongly Agree, with Strongly Agree alone accounting for 43.1%; only 7.0% of answers fall on the disagree side. The grand mean is 4.138 on the 1–5 scale, i.e. +1.14 on the $[-2, +2]$ encoding.

Two consequences follow, and both shape every later decision. Item means barely discriminate between statements, so the informative signal is in which statements attract disagreement rather than in overall agreement level. And because almost every respondent agrees with almost everything, two respondents will look similar under a raw similarity measure simply by both being agreeable. Section 3.3 removes this component explicitly.

Figure 1(b) shows the ceiling is not uniform across topics. Environment ($\mu = 4.39$, 89.5% agree) and society ($\mu = 4.28$, 86.0%) are close to settled consensus, whereas technology ($\mu = 3.99$, 8.1% disagree) and especially education ($\mu = 3.91$, 13.8% disagree) retain real disagreement. Notably, the blocks with the most disagreement are also those with the lowest internal consistency, consistent with their items tapping several distinct opinions rather than one settled disposition.

2.2 Where the disagreement lives

Figure 2 ranks the twelve items attracting the most disagreement. Seven are education items and four are technology items; only one comes from society, and none from environment. Compulsory class attendance (E03, mean 2.06) is the only statement the cohort actively rejects, with 70.1% disagreeing and 40% strongly disagreeing. Written examinations as a measure of knowledge (E02, mean 2.84) is the next most rejected.

The pattern is specific rather than generally oppositional: the same cohort returns 94.3% agreement that curricula should be updated more often (E11) and that universities should prioritise problem-solving over rote learning (E10). What is contested is assessment and teaching format, not the institution.

The technology items behave similarly. Teaching AI (T04, 81.3%), AI as a learning aid (T02, 79.1%), and AI accelerating research (T05, 81.3%) all score high, whereas AI-assisted medical diagnosis (T08, mean 3.08) and autonomous vehicles (T09, mean 3.28) are the two lowest technology items by a clear margin. Since robots in hazardous occupations (T07) scores 82.4% agreement, the objection is not to automation in general but to delegating consequential judgment.

2.3 Missingness and data-quality notes

Of the 96 records, 85 are complete, 5 are entirely empty, 4 stop after the technology block, and 2 stop partway through environment. Dropout is therefore monotone—a truncated tail rather than scattered gaps—which matters because it means later blocks rest on a slightly smaller base ($n = 84$ – 91) than earlier ones. Separately, 37 cells carry No Comments, scattered across 17 respondents who otherwise completed the questionnaire.

Three limitations should be carried into every result below. Items with means above roughly 4.5 measure social-desirability agreement more than strength of preference. Only a small minority of statements are worded to invite disagreement from an otherwise positive respondent, so acquiescence bias and a genuine ceiling cannot be fully separated here. And the file contains no demographic fields, so no finding can be attributed to a subgroup.

3 Data Handling

3.1 Dataset sanitation

Missingness structure. Five entirely empty records are excluded, leaving 91 respondents. No respondent is excluded for being merely incomplete: the four who stopped after technology and the two who stopped during environment are retained, and their unanswered items are treated as missing data to be recovered rather than as grounds for deletion. This yields 205 contiguous structural blank cells and 37 NC cells, 242 missing cells in total among the retained respondents.

We distinguish

structural blank = question not reached,

from

NC = deliberate non-response.

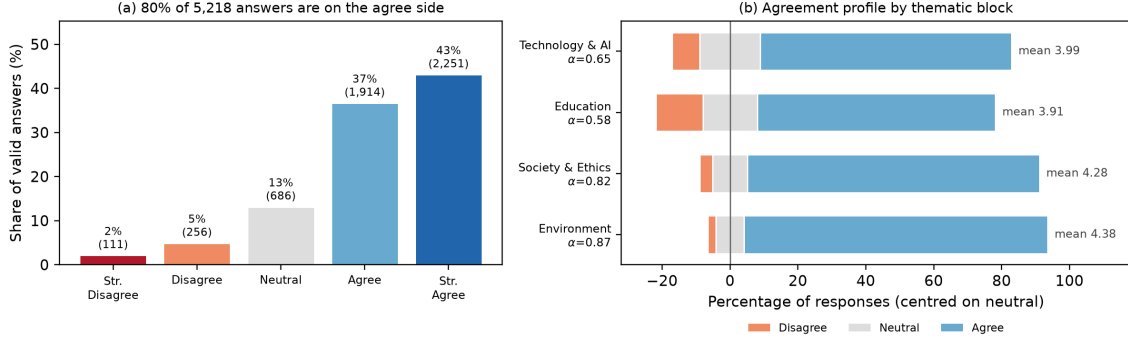


Figure 1: The response ceiling. (a) Distribution of all 5,218 valid answers: nearly four in five are on the agree side and 43% are Strongly Agree. (b) Agreement profile per block, centred on neutral, with block mean and Cronbach’s α . Environment and society are near consensus; education carries the most disagreement and the weakest internal consistency.

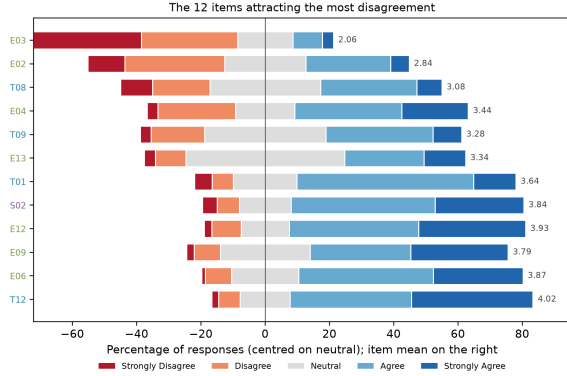


Figure 2: The twelve items attracting the most disagreement, as diverging response profiles centred on neutral, with item mean on the right. Labels are coloured by block. Disagreement concentrates in education (7 of 12) and technology (4 of 12).

Both become missing values during modelling, but their original types are retained so that preprocessing never silently conflates abandonment with an explicit decision to withhold an answer.

Why NC is not Neutral. Recoding NC as Neutral would assign a substantive response to a cell for which none was supplied. As a diagnostic we fitted a two-way additive model with respondent and item effects on the clean responses, $\hat{x}_{iq} = \bar{\mu} + a_i + b_q$. The fitted value at the 37 NC cells averaged 3.92 on the 1–5 scale against 3 for Neutral, so recoding would impose a systematic downward shift of about 0.92 scale points. This model is used only as a diagnostic, not as the imputation method.

Table 1: Sensitivity of the item-correlation network to NC treatment at $\tau = 0.30$, against pairwise deletion as reference. Computed at an earlier 86-respondent configuration; ordering of the treatments is the result of interest rather than the absolute counts.

Treatment	N	E	r	Jacc.	AMI
Pairwise delete (ref.)	86	325	—	—	—
Recode $\rightarrow$ Neutral	86	333	.993	.875	.662
Drop NC users	69	362	.947	.616	.337
Two-way impute	86	335	.997	.924	.670

Empirical NC diagnostics. NC usage shows no systematic item- or respondent-level structure under the tests applied. NC counts per item have a dispersion index of 1.10 and a chi-squared test of item-level uniformity gives $p = 0.27$; no item carries more than three NC responses; the correlation between NC frequency and item standard deviation is $r = -0.02$. At respondent level, NC users have mean score 4.01 against 4.15 for non-users ($p = 0.27$), and neutral-response rates of 0.123 against 0.142 ($p = 0.82$). There is therefore no empirical basis for assigning NC a fixed ordinal category, and we retain it as unobserved.

Sensitivity to NC treatment. Table 1 compares four NC treatments on the item-correlation network at $\tau = 0.30$. Dropping NC users would remove 17 respondents, roughly a fifth of the sample, to eliminate only 37 cells, and it is the only treatment that materially disturbs the recovered community structure (AMI 0.337). Correlations are far more stable than partitions across all four treatments—a warning that stability of a correlation matrix should never be read as evidence that a particular partition of it is stable.

Table 2: Artificial-masking validation. Accuracy is exact-category; MAE is mean absolute error on the five-point scale.

Masking	Ordinal model		Mode baseline	
	Acc.	MAE	Acc.	MAE
Scattered (NC-like)	.539	.534	.486	.626
Suffix (structural)	.498	.612	.482	.655

3.2 Missing-value imputation

Ordinal regression formulation. For target item j the remaining 59 items form the predictor vector $\mathbf{x}_{i,-j}$, and the target satisfies $y_{ij} \in \{-2, -1, 0, +1, +2\}$. We fit one regularised proportional-odds ordinal logistic regression [6] per item, with latent score $z_i = \beta_0 + \beta^\top \mathbf{x}_{i,-j}$ and ordered thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$ inducing the five response intervals. The prediction is the maximum-probability category,

$$\hat{y}_{ij} = \arg \max_k P(y_{ij} = k \mid \mathbf{x}_{i,-j}).$$

This preserves the ordering of the Likert responses while avoiding the interval-scale assumption of linear regression, and unlike nominal multiclass classification it retains the relative order of the five outcomes.

Missing predictors and regularisation. A respondent may have several missing cells, so predictors that are themselves missing cannot be encoded as Neutral. They are instead filled with the training-fold column median before standardisation, with the statistic computed only from training data in each fold to prevent leakage from held-out responses. A ridge penalty controls complexity in the 59-dimensional predictor space, and the target is always excluded from its own predictor vector, giving $M_j : \mathbf{X}_{-j} \rightarrow Y_j$ for $j = 1, \dots, 60$.

Validation. The true values of the 242 genuinely missing cells are unavailable, so the pipeline is evaluated by artificial masking of responses whose values are known, under two mechanisms matched to the two missingness types. Scattered masking hides 37 observed cells at random inside otherwise complete respondents, approximating NC, over 40 trials. Suffix masking hides a contiguous tail after a simulated stopping point drawn from the observed stopping pattern, for 15 respondents per trial over 15 trials; items after the stopping point are withheld from the predictor so the model cannot use information unavailable in practice.

Because the target is ordinal, MAE is the more informative metric: a prediction one category away is not the same error as one at the opposite end of the scale. Table 2 shows the model beats a modal-response baseline under both mechanisms, by a wider margin

under scattered masking where more co-answered items remain available. Performance is lower under suffix masking precisely because fewer predictors survive the stopping point, so the two settings should be read separately rather than averaged into one number. The final model imputes all 242 missing cells, with every recovered value restricted to the original five-level support.

3.3 Row-centring

Imputation yields a complete 91×60 matrix on the original scale, but not one that can be correlated directly. Because the grand mean is +1.14 (Section 2), a respondent who agrees with everything raises all 60 of their entries together, which appears as association between every pair of items at once. We therefore replace each entry by its deviation from that respondent’s own mean,

$$x'_{i,j} = x_{i,j} - \bar{x}_i, \quad \bar{x}_i = \frac{1}{60} \sum_{k=1}^{60} x_{i,k}. \quad (1)$$

Two points are worth stating precisely. First, centring is applied strictly after imputation and strictly before correlation: the imputation model predicts on the original -2 to $+2$ scale, and centring is a network-construction step, not a data-cleaning one. Second, rank correlation does not remove this bias automatically—Spearman correlation is invariant to a monotone transform of a single column, not to an additive shift that differs from row to row. The magnitude of the effect is reported as Graph 3 (Section 4.3).

3.4 Network construction

On the sanitised, row-centred matrix we build a weighted undirected graph $G = (V, E, W)$ whose 60 vertices are the questionnaire items. Because responses are ordered categories, the primary association measure is the Spearman rank correlation ρ_{qr} , and edge weights are signed correlations, with $|\rho_{qr}|$ used for edge magnitude in visualisation and sign distinguished by colour. We retain the full weighted matrix and threshold only for display, showing an edge when $|\rho_{qr}| \geq \tau$. The association step is implemented modularly so a latent-variable ordinal measure such as polychoric correlation can replace Spearman without changing graph construction. A second, respondent-level network is introduced in Section 4.12 for the dynamics experiment.

4 Analysis and Visualisations

This section reports the twelve analyses. Each states what the figure shows, what was found, and what fol-

lows from it. Sections still owned by other team members are marked as placeholders.

A note on respondent counts. Several reference values circulated during planning assume a pipeline variant that drops respondents with more than 12 missing items, leaving 86. This report does not use that filter: all 91 non-empty respondents are retained and their missing cells recovered by ordinal regression (Section 3.2). Figures produced here therefore report $n = 91$, and small deviations from those reference numbers are expected.

4.1 Graph 1: Missingness map and response distribution

Figure 3 shows every cell of the raw 96×60 matrix as missing or present, with respondents sorted by missing count, beside the distribution of per-respondent mean scores.

Findings. The 542 missing cells (505 structural blanks plus 37 NC) are extremely unevenly distributed. Five respondents are entirely empty and account for 300 of them; the remaining 242 are spread thinly across the other 91, a missing rate of 4.4%. Below the cut-off the map is almost clean, with scattered single cells and two visible tails from the respondents who stopped early. The right panel locates the grand mean at 1.14, with the bulk of respondents between 0.75 and 1.50 and only one respondent below zero.

Implication. The missingness structure justifies the retention policy directly. Because all but five respondents answered at least 80% of items, there is no population of badly-incomplete respondents whose removal would simplify the analysis, and a count-based exclusion rule would discard usable respondents to remove very few cells. The right panel is the first direct evidence of the acquiescence problem that Section 3.3 corrects.

4.2 Graph 2: Item variance ranking

Figure 4 ranks all 60 items by standard deviation on the uncentred imputed matrix. Centring is deliberately not applied here: it changes each item’s variance and would answer a different question (spread after removing each respondent’s baseline) than the one asked (how divided the cohort is on the statement as posed).

Findings. Item SD ranges from 0.50 (E15, “continuous learning is essential”, near-unanimous) to 1.18 (E04, online learning). The five most contested items are E04, E03, E02, T08 and T07; the least contested include V13, V06, S05 and E15. By block mean SD the order is technology 0.92, education 0.88, society 0.79, environment 0.72.

Implication. The block averages and the extremes tell slightly different stories, and both matter. Technology has the highest average variance, but the sharpest disagreements are educational: three of the five most divided items are E02–E04, all concerning assessment and teaching format. Environment contributes none of the top ten. This is a property of the survey instrument, not of preprocessing: the questionnaire contains a settled domain (environment), a moderately settled one (society), and two live ones (technology, education). Consensus items carry little information for a correlation network—an item everyone endorses cannot covary with anything—so the network’s signal is necessarily concentrated in the contested tail.

4.3 Graph 3: The effect of row-centring

Figure 5 overlays the distribution of all $\binom{60}{2} = 1,770$ pairwise item correlations computed before and after row-centring, on identical bin edges.

Findings. The raw distribution is displaced positive, with mean $r = 0.151$ and median 0.151; almost no pair is appreciably negative. After centring the distribution sits essentially on zero, mean $r = -0.004$, median -0.008 , with positive skew (0.221) leaving a heavier right tail of genuinely co-varying pairs. At a fixed threshold of $|r| \geq 0.30$ the number of surviving pairs falls from **327** to **42**, a reduction of 87%.

Implication. This converts the claim of Section 3.3 from an assertion into a measurement: roughly seven eighths of the apparent association in the raw matrix is shared response style, not item relationship. Had the network been built on raw scores it would have been dense, almost entirely positive, and would have supported confident but spurious conclusions about item structure. The surviving 42 pairs are the only ones worth interpreting, and the emergence of negative correlations after centring—impossible to see in the raw distribution—is what later permits a signed-network analysis at all.

4.4 Graph 4: Real versus shuffled correlations

[TO BE COMPLETED — Graph 4 — owner: P2]

Compare the distribution of observed item–item correlations against a column-shuffled null that destroys inter-item structure while preserving each item’s marginal. Plot both distributions and report the excess of real edges over chance at several thresholds.

Reference values to check against: real SD 0.136 vs shuffled SD 0.108 (theoretical $1/\sqrt{n-3} \approx 0.109$); edges at $|r| > 0.15$: 467 real vs 309 chance; > 0.20 : 258 vs 118; > 0.25 : 111 vs 37; > 0.30 : 49 vs 9.

Findings + implication: to be written.

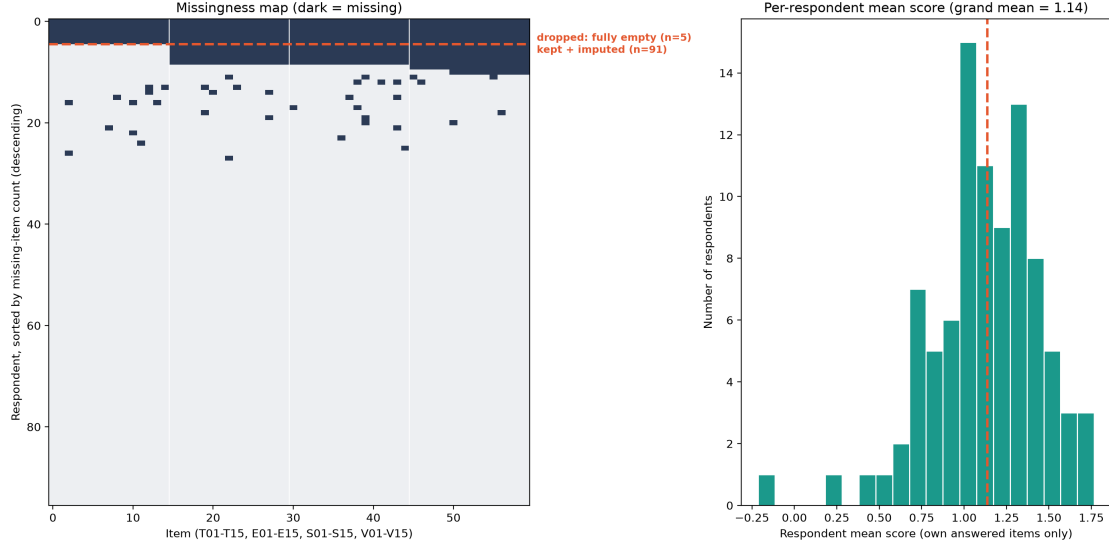


Figure 3: Graph 1. Left: missingness map over all 96 raw respondents, sorted by missing-item count; dark cells are missing for either reason. The dashed line marks this project's exclusion rule—only the five entirely empty records are dropped, and the remaining 91 respondents have their 242 missing cells imputed. Right: per-respondent mean over answered items only, with the grand mean at 1.14.

4.5 Graph 5: Eigenvalue spectrum against Marchenko–Pastur

[TO BE COMPLETED — Graph 5 — owner: P2]

Plot the eigenvalue spectrum of the row-centred correlation matrix against the Marchenko–Pastur band [4] for a random matrix of the same shape, to separate informative modes from noise.

Reference values: with T respondents and $N = 60$ items, $Q = T/N$ and $\lambda_+ = (1 + 1/\sqrt{Q})^2$. At $T = 86$ this gives $\lambda_+ \approx 3.368$, with exactly three eigenvalues above it (4.05, 3.98, 3.53) carrying 19.3% of total variance. Recompute for $T = 91$.

Findings + implication: to be written. Note that PC1 explains only about 8% of variance, so this should not be read as evidence of a two-camp polarisation.

4.6 Graph 6: Reordered correlation heatmap

[TO BE COMPLETED — Graph 6 — owner: P3]

Correlation matrix with rows and columns reordered by detected community so that communities appear as diagonal blocks, with a colour scale that distinguishes sign.

Findings + implication: to be written. State explicitly whether the diagonal blocks align with the four thematic blocks or cut across them — that comparison is the point of the figure.

4.7 Graph 7: Threshold and percolation sweep

[TO BE COMPLETED — Graph 7 — owner: P2]

Sweep the edge threshold τ and plot number of connected components, modularity, and average clustering against it, to select τ by evidence rather than convention. This figure supplies the threshold used by Graph 8.

Findings + implication: to be written. Report the chosen τ and the justification, and note the trade-off between a connected graph and one whose edges are individually trust-worthy.

4.8 Graph 8: Main item network

[TO BE COMPLETED — Graph 8 — owner: P1, blocked on Graph 7]

The item–item network at the threshold selected in Graph 7, nodes coloured by thematic block, edges signed and weighted by correlation magnitude.

Dependency: the threshold must come from the Graph 7 sweep rather than being chosen here.

Findings + implication: to be written. Report the number of connected items and isolates at the chosen threshold, which items act as hubs, and whether edges respect block boundaries.

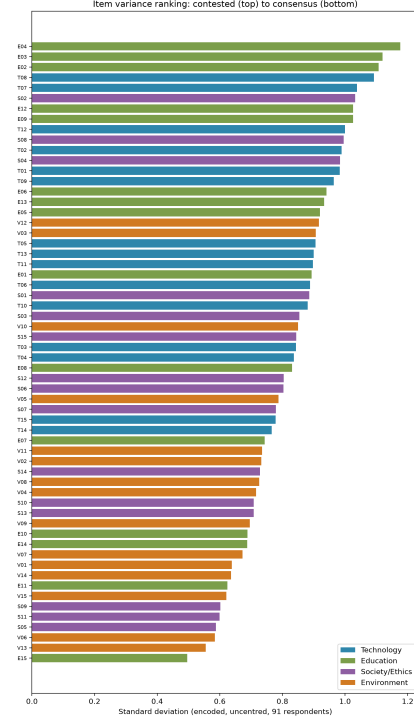


Figure 4: Graph 2. All 60 items ranked by standard deviation on the uncentred imputed matrix, coloured by block. Contested items (top) concentrate in education and technology; environment items cluster at the consensus end.

4.9 Graph 9: Topic-block connectivity

[TO BE COMPLETED — Graph 9 — owner: P1]

4 × 4 heatmap of connectivity between the four thematic blocks, summarising how strongly each pair of topics is linked.

Reference values (respondent-level block-score correlations): T-E 0.29, T-S 0.54, T-V 0.46, E-S 0.53, E-V 0.49, S-V **0.70**. The raw values are available in data/uc_survey_derived.json.

Findings + implication: to be written. The headline is that society and environment behave as a single “civic and ecological responsibility” disposition, while technology and education are close to independent ($r = 0.29$, the weakest pair) — views on AI and views on pedagogy are separate axes.

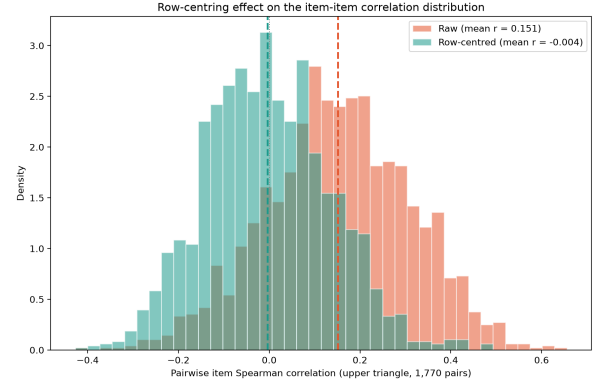


Figure 5: Graph 3. Distribution of all 1,770 pairwise item correlations before and after row-centring, on shared bin edges. Centring moves the mean from 0.151 to -0.004 and removes 87% of the pairs exceeding $|r| \geq 0.30$.

4.10 Graph 10: Communities against null models

[TO BE COMPLETED — Graph 10 — owner: P3]

Compare observed modularity and clustering against three nulls: a degree-preserving rewiring [5], an Erdős-Rényi graph, and a column-permutation null that regenerates the entire pipeline from scrambled responses.

Reference values (respondent network): modularity 0.408–0.423 and clustering 0.202 observed; rewiring null 0.329 ± 0.011 ($z \approx 7$) and 0.091 ± 0.013 ($z \approx 8.9$), both passed; permutation null 0.403 ± 0.017 ($z \approx 1.15$), not passed.

Findings + implication: to be written. The permutation null is the decisive one: rewiring and ER take the finished graph as given and cannot detect transitivity introduced by the correlation-and-threshold construction itself. If modularity clears rewiring but not permutation, the community structure is largely a construction artefact and must be reported as such — this is a finding, not a failure.

4.11 Graph 11: Structural balance

[TO BE COMPLETED — Graph 11 — owner: P3]

Fraction of balanced triads in the signed item network as a function of threshold, against a sign-permutation null [1].

Reference values: balanced-triad fraction 83% at $|r| > 0.15$, 96% at 0.20, 100% at 0.25, against roughly 50% under the null.

Findings + implication: to be written. That the fraction strengthens as the threshold tightens is itself part of the evidence, since a spurious effect would not sharpen as weak edges are removed.

4.12 Graph 12: Opinion dynamics

This is the only analysis involving an actual dynamical process; everything above describes a fixed dataset. Because each opinion is one respondent’s answer to

one item, the vertices here are respondents, not items, so a second network is required.

The respondent network. From the row-centred matrix we compute the 91×91 Pearson correlation between respondents and connect each respondent to their $k = 4$ most similar peers, symmetrising by union. Union rather than mutual k NN is deliberate: mutual k NN fragments this data badly, and disconnected components cannot merge at any confidence threshold, which would inflate the final cluster count for reasons unrelated to the dynamics. The result has 91 nodes, 298 edges, mean degree 6.55, and forms a single connected component.

The model. Opinions $x_i \in [0, 1]$ are seeded from item E03 (compulsory attendance, SD 1.12), rescaled by $s(x) = (x + 2)/4$. At each step a random edge (u, v) is drawn, and if the two opinions lie within the confidence threshold they move to their midpoint [3]:

$$|x_u - x_v| < \varepsilon \implies x_u, x_v \leftarrow \frac{1}{2}(x_u + x_v). \quad (2)$$

We sweep ε from 0.05 to 0.60 in steps of 0.05, with 20 seeded repetitions of up to 60,000 interactions each, terminating early on convergence, and count final clusters by rounding to two decimals.

Findings. Results appear in Figure 6 and Table 3. Cluster count is exactly 5.00 ± 0.00 for every $\varepsilon \leq 0.25$, then falls to 4.45 at $\varepsilon = 0.30$, 3.35 at 0.40, 2.00 at 0.50, approaching consensus near 0.55–0.60.

The plateau is an artefact, and provably so. The specification for this analysis requires stating that the low- ε plateau is partly an artefact of the discrete five-point grid. In fact the claim can be made exactly. Under $s(x) = (x + 2)/4$ the five categories map to $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, so any two distinct initial opinions differ by at least $\frac{1}{4}$. For any $\varepsilon \leq 0.25$ the condition $|x_u - x_v| < \varepsilon$ is therefore never satisfied by a pair with distinct opinions: the initial state is a fixed point and the final count equals the number of occupied categories, on any graph whatsoever.

Three observations confirm this empirically. The plateau has zero variance across 20 randomly seeded runs, which no genuine emergent outcome would. A complete graph—a maximally different topology—returns the identical 5.00 ± 0.00 over the same range. And re-running with seeds jittered by ± 0.125 , which destroys the grid while preserving the distribution’s shape, removes the plateau entirely, yielding $35.95 \rightarrow 26.35 \rightarrow 21.05 \rightarrow 14.20 \rightarrow 8.95$ clusters across $\varepsilon = 0.05 \dots 0.25$. The five clusters below $\varepsilon = 0.25$ are the original response categories failing to merge, not opinion groups forming, and the break in the curve at $\varepsilon = 0.30$ is arithmetic rather than a discovered threshold.

Table 3: Deffuant–Weisbuch final cluster counts (mean $\pm$ sd over 20 repetitions). The plateau at $\varepsilon \leq 0.25$ is forced by the response grid and appears identically on any topology; the gap between the k NN and complete graphs above it is the genuine topological effect.

ε	k NN graph	Complete graph	Jittered control
≤ 0.25	5.00 ± 0.00	5.00 ± 0.00	$36.0 \rightarrow 9.0$
0.30	4.45 ± 1.12	2.45 ± 0.50	6.50 ± 1.60
0.35	4.00 ± 1.55	2.00 ± 0.00	5.10 ± 1.09
0.40	3.35 ± 0.79	2.00 ± 0.00	3.95 ± 1.28
0.45	3.15 ± 0.85	2.25 ± 0.43	2.95 ± 0.92
0.50	2.00 ± 0.77	2.05 ± 0.22	2.35 ± 0.79
0.55	1.65 ± 0.48	1.95 ± 0.22	1.65 ± 0.48
0.60	1.30 ± 0.46	1.65 ± 0.48	1.20 ± 0.40

What is not an artefact. The substantive result lies above $\varepsilon = 0.25$. Comparing the real network against an identically seeded complete graph—the same dynamics with topology removed—the well-mixed population collapses to about 2 clusters across $\varepsilon = 0.30$ –0.45, while the sparse k NN network sustains 3.15–4.45. Network structure therefore sustains opinion diversity that a well-mixed population loses at the same tolerance: when respondents interact only with their nearest neighbours in opinion space, locally cohesive groups stabilise before global consensus can propagate. This is the one result in the figure that is neither a scale artefact nor a restatement of the input distribution, and it is what justifies studying the process on a network at all.

5 Results and Discussion

5.1 What the data supports

Superficial consensus with genuine local structure. The headline finding is not polarisation. Agreement runs at 79.8% across every block and the leading principal component explains only about 8% of variance, so there is no evidence of two opposed camps. What exists instead is a broad agreeable consensus with real but local disagreement concentrated in specific, identifiable places.

Disagreement is domain-specific, not dispositional. Environment and society are effectively settled; technology and education are not. Within education the cohort is precise about what it rejects—compulsory attendance and written examinations—while endorsing curriculum reform and problem-solving at above 94%. Within technology the objection tracks delegated consequential judgment (medical diagnosis, autonomous vehicles) rather than automation as such.

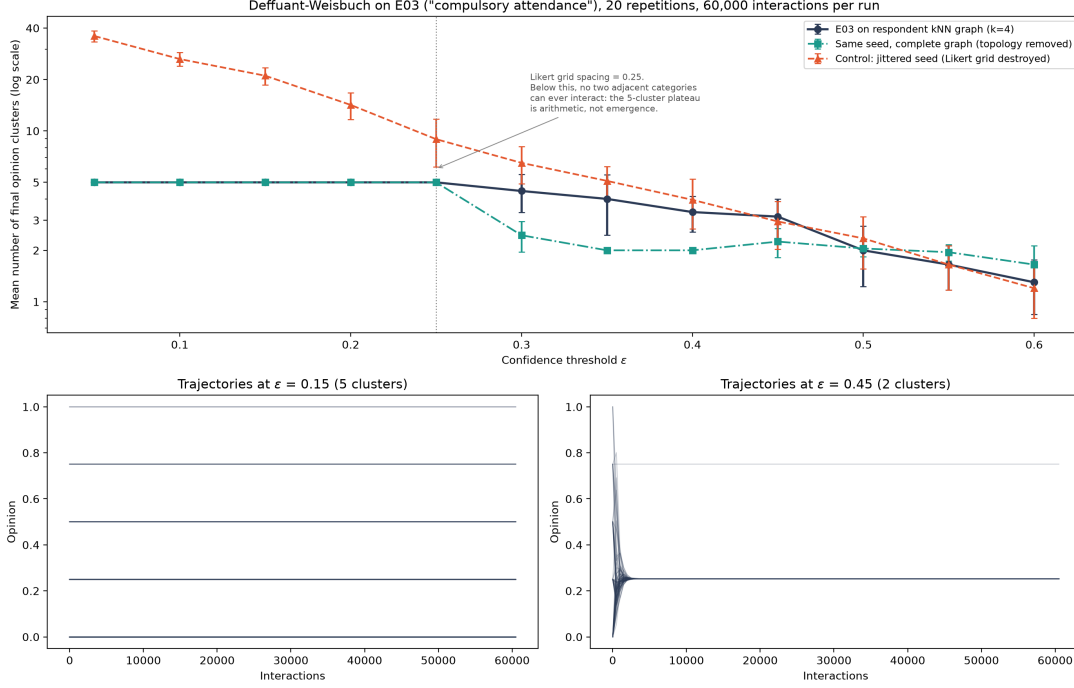


Figure 6: Graph 12. Top: mean final cluster count against confidence threshold ϵ on a log scale, with error bars over 20 repetitions. The k NN network and the complete graph coincide exactly below $\epsilon = 0.25$ —the signature of an arithmetic artefact—and diverge above it, where topology matters. The jittered control destroys the response grid and removes the plateau entirely. Bottom: individual opinion trajectories at a low and a high ϵ , showing frozen categories against rapid collapse.

Most apparent structure was response style. Row-centring removes 87% of the item pairs exceeding $|r| \geq 0.30$. Any analysis of this questionnaire that skips this step will find dense, uniformly positive structure and will be measuring agreeableness.

Topology has a measurable dynamical consequence. In the bounded-confidence model, the sparse respondent network sustains noticeably more opinion clusters than a well-mixed population at the same tolerance, over the range where the dynamics are genuinely active.

5.2 What the data does not support

[TO BE COMPLETED — Results and Discussion — integration, owner: P3]

Once Graphs 4–11 are complete, integrate their findings here. In particular this section must state:

- (i) whether the observed community structure survives the permutation null of Graph 10, and if it does not, report that the communities are largely a construction artefact — as a finding;
- (ii) how many eigenvalue modes exceed the Marchenko–Pastur band (Graph 5) and what fraction of variance is therefore attributable to signal rather than noise;
- (iii) whether the detected communities align with the four thematic blocks or cut across them (Graphs 6 and 9);
- (iv) whether structural balance strengthens with threshold (Graph 11).

5.3 Methods considered and rejected

Documenting what did not work is part of the analysis. Four alternatives were evaluated and set aside.

Recoding NC as Neutral imposes a systematic shift of about 0.92 scale points at those cells and asserts a response that was explicitly withheld. Dropping NC users removes roughly a fifth of the sample to eliminate 37 cells and is the only NC treatment that sub-

stantially disturbs the recovered partition (AMI 0.337). Excluding respondents by missing-item count was rejected because, once the five empty records are removed, no substantial population of badly-incomplete respondents remains: the rest are 95.6% complete, and model-based recovery is preferable to discarding usable respondents. Mutual k NN for the respondent network fragments the graph into many components, which would inflate cluster counts in the dynamics for reasons unrelated to the process being studied; union k NN yields a single connected component.

[TO BE COMPLETED — Rejected methods — additions from P2/P3]

Add the rejected sparsification and similarity variants once measured: percentile thresholding at p90 (reported to leave 22 components and many isolates), cosine similarity and Gower distance as alternatives to Pearson, and raw uncentred similarity (near-complete graph — partially covered by Graph 3).

5.4 Limitations

Four limits bound every claim above. The response ceiling means items with means above roughly 4.5 measure social desirability more than preference strength, and because few items are reverse-worded, acquiescence and a genuine ceiling cannot be fully separated. The sample is $n = 96$ from a single institution, so differences of a few percentage points between statements are noise and only the large gaps are safely above sampling variation. Dropout is monotone, so later blocks rest on a slightly smaller base than earlier ones. And the file carries no demographic fields, so no finding can be attributed to any subgroup without additional joined data.

A methodological limitation deserves separate mention: correlation stability under perturbation does not imply partition stability. Table 1 shows correlation matrices agreeing at $r > 0.94$ across NC treatments whose community assignments agree at AMI 0.337. Community level conclusions require their own stability analysis, by respondent-level bootstrap resampling, before being treated as findings.

6 Conclusion

We presented an end-to-end pipeline from a raw ordinal questionnaire to an opinion network and a dynamical process on it. Sanitisation preserves the distinction between abandonment and deliberate non-response, retains all 91 non-empty respondents, and recovers 242 missing cells by question-wise regularised proportional-odds ordinal regression, which beats a modal baseline under both scattered and structural artificial masking. Row-centring then removes the acquiescence component that would otherwise dominate every correla-

tion, an intervention whose magnitude we quantify rather than assume: it eliminates 87% of item pairs correlated at $|r| \geq 0.30$. On the resulting matrix we build signed item and respondent networks and run bounded-confidence dynamics, showing by direct argument and controlled experiment that the small- ε cluster plateau is forced by the five-point response grid, while the diversity sustained by the sparse network relative to a well-mixed population is a genuine topological effect.

[TO BE COMPLETED — Conclusion — final integration]

Complete once Graphs 4–11 land. State the final chosen threshold and its justification, the number of communities and whether they survive the permutation null, and close on the overall characterisation of the cohort — superficial consensus with genuine local clustering — supported by the completed null-model evidence rather than asserted.

Individual Contribution

Table 4: Contribution by team member.

Member	Contribution
Arijeet Paul	Data sanitation and missingness analysis; ordinal-regression imputation and its masking validation; row-centring; Graphs 1, 2, 3 and 12; respondent k NN network; repository structure and run_all.py.
Shreyash Lohare	[TO BE COMPLETED — Graphs 4, 5, 7]
Dev Patel	[TO BE COMPLETED — Graphs 6, 10, 11]

[TO BE COMPLETED — Individual contribution]

Each member should replace their row with specific, auditable contributions (modules written, figures produced, sections drafted). git log provides the record.

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